

Pipelining Problems

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Speedup formula

- $$\text{Speedup} = \frac{\text{Time taken for non pipelined (tn)}}{\text{Time taken for pipelined version(tp)}}$$

t_n = summation of all timings

$T_p = \max\{t_i\} + \text{delay latch}$

Problem 1

Draw a space-time diagram for a six-segment pipeline showing the time it takes to process eight tasks.

Segment	1	2	3	4	5	6	7	8	9	10	11	12	13
1	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8					
2		T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8				
3			T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8			
4				T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8		
5					T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8	
6						T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8

$$(k - n - 1)t_p = 6 + 8 - 1 = 13 \text{ cycles}$$

Problem 2

Determine the number of clock cycles that it takes to process 200 tasks in a six-segment pipeline.

$$\begin{array}{l} k = 6 \text{ segment} \\ n = 200 \text{ tasks} \end{array} \quad (k + n - 1) = 6 + 200 - 1 = 205 \text{ cycles}$$

Problem 3

Consider a pipeline having 4 phases with duration 60, 50, 90 and 80 ns. Given latch delay is 10 ns. Calculate-

1. Pipeline cycle time
2. Non-pipeline execution time
3. Speed up ratio
4. Pipeline time for 1000 tasks
5. Sequential time for 1000 tasks
6. Throughput

Cycle time

= Maximum delay due to any stage + Delay due to its register

= $\text{Max} \{ 60, 50, 90, 80 \} + 10 \text{ ns}$

= 90 ns + 10 ns

= 100 ns

Non-pipeline execution time for one instruction

$$= 60 \text{ ns} + 50 \text{ ns} + 90 \text{ ns} + 80 \text{ ns}$$

$$= 280 \text{ ns}$$

Speed up

$$= \text{Non-pipeline execution time} / \text{Pipeline execution time}$$

$$= 280 \text{ ns} / \text{Cycle time}$$

$$= 280 \text{ ns} / 100 \text{ ns}$$

$$= 2.8$$

Pipeline time for 1000 tasks

$$= \text{Time taken for 1st task} + \text{Time taken for remaining 999 tasks}$$

$$= 1 \times 4 \text{ clock cycles} + 999 \times 1 \text{ clock cycle}$$

$$= 4 \times \text{cycle time} + 999 \times \text{cycle time}$$

$$= 4 \times 100 \text{ ns} + 999 \times 100 \text{ ns}$$

$$= 400 \text{ ns} + 99900 \text{ ns}$$

$$= 100300 \text{ ns}$$

Non-pipeline time for 1000 tasks

= 1000 x Time taken for one task

= 1000 x 280 ns

= 280000 ns

Throughput for pipelined execution

= Number of instructions executed per unit time

= 1000 tasks / 100300 ns

Problem 4

A four stage pipeline has the stage delays as 150, 120, 160 and 140 ns respectively. Registers are used between the stages and have a delay of 5 ns each. Assuming constant clocking rate, the total time taken to process 1000 data items on the pipeline will be?

Cycle time

= Maximum delay due to any stage + Delay due to its register

= $\text{Max} \{ 150, 120, 160, 140 \} + 5 \text{ ns}$

= $160 \text{ ns} + 5 \text{ ns}$

= 165 ns

Pipeline time to process 1000 data items

= Time taken for 1st data item + Time taken for remaining 999 data items

= $1 \times 4 \text{ clock cycles} + 999 \times 1 \text{ clock cycle}$

= $4 \times \text{cycle time} + 999 \times \text{cycle time}$

= $4 \times 165 \text{ ns} + 999 \times 165 \text{ ns}$

= $660 \text{ ns} + 164835 \text{ ns}$

= 165495 ns

= $165.5 \mu\text{s}$

Problem 5

Consider a non-pipelined processor with a clock rate of 2.5 gigahertz and average cycles per instruction of 4. The same processor is upgraded to a pipelined processor with five stages but due to the internal pipeline delay, the clock speed is reduced to 2 gigahertz. Assume there are no stalls in the pipeline. The speed up achieved in this pipelined processor is?

Cycle time in non-pipelined processor

Frequency of the clock = 2.5 gigahertz

Cycle time

$$= 1 / \text{frequency}$$

$$= 1 / (2.5 \text{ gigahertz})$$

$$= 1 / (2.5 \times 10^9 \text{ hertz})$$

$$= 0.4 \text{ ns}$$

Non-pipeline execution time to process 1 instruction

= Number of clock cycles taken to execute one instruction

= 4 clock cycles

$$= 4 \times 0.4 \text{ ns}$$

$$= 1.6 \text{ ns}$$

Cycle time of pipelined processor

Frequency of the clock = 2 gigahertz

Cycle time

= $1 / \text{frequency}$

= $1 / (2 \text{ gigahertz})$

= $1 / (2 \times 10^9 \text{ hertz})$

= 0.5 ns

Pipeline Execution Time

Since there are no stalls in the pipeline, so ideally one instruction is executed per clock cycle. So,

Pipeline execution time

= 1 clock cycle

= 0.5 ns

Speed up

= Non-pipeline execution time / Pipeline execution time

= 1.6 ns / 0.5 ns

= 3.2

Problem 6

The stage delays in a 4 stage pipeline are 800, 500, 400 and 300 picoseconds. The first stage is replaced with a functionally equivalent design involving two stages with respective delays 600 and 350 picoseconds.

The throughput increase of the pipeline is _____%.

Cycle time

= Maximum delay due to any stage + Delay due to its register

= $\text{Max} \{ 800, 500, 400, 300 \} + 0$

= 800 picoseconds

Thus, Execution time in 4 stage pipeline = 1 clock cycle = 800 picoseconds.

Throughput

= Number of instructions executed per unit time

= 1 instruction / 800 picoseconds

Cycle time

= Maximum delay due to any stage + Delay due to its register

= $\text{Max} \{ 600, 350 \} + 0$

= 600 picoseconds

Thus, Execution time in 2 stage pipeline = 1 clock cycle = 600 picoseconds.

Throughput

= Number of instructions executed per unit time

= 1 instruction / 600 picoseconds

Throughput increase

= $\{ (\text{Final throughput} - \text{Initial throughput}) / \text{Initial throughput} \} \times 100$

= $\{ (1 / 600 - 1 / 800) / (1 / 800) \} \times 100$

= $\{ (800 / 600) - 1 \} \times 100$

= $(1.33 - 1) \times 100$

= 0.3333×100

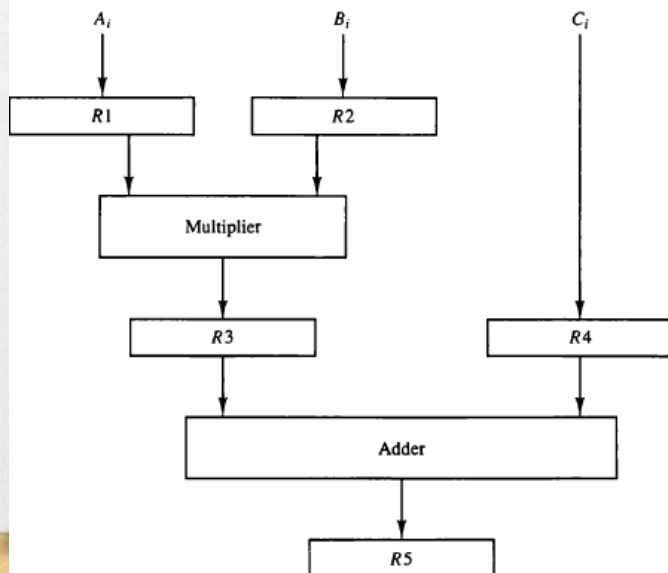
= 33.33 %

Problem 7

The pipeline of Fig. 9-2 has the following propagation times: 40 ns for the operands to be read from memory into registers R1 and R2, 45 ns for the signal to propagate through the multiplier, 5 ns for the transfer into R3, and 15 ns to add the two numbers into R5.

- What is the minimum clock cycle time that can be used?
- A nonpipeline system can perform the same operation by removing R3 and R4. How long will it take to multiply and add the operands without using the pipeline?
- Calculate the speedup of the pipeline for 10 tasks and again for 100 tasks.
- What is the maximum speedup that can be achieved?

Figure 9-2 Example of pipeline processing.



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- What is the maximum speedup that can be achieved?

$$(a) t_p = 45 + 5 = 50 \text{ ns} \quad k = 3$$

$$(b) t_n = 40 + 45 + 15 = 100 \text{ ns}$$

$$(c) S = \frac{n t_n}{(k+n-1) t_p} = \frac{10 \times 100}{(3+9) 50} = 1.67 \quad \text{for } n = 10$$

$$= \frac{100 \times 100}{(3+99) 50} = 1.96 \quad \text{for } n = 100$$

$$(d) S_{\max} = \frac{t_n}{t_p} = \frac{100}{50} = 2$$